

Understanding the Sleep Apnea .mat Data

A practical exploration plan before model development

342,540 x 7	342,540	4.76 hours	20.00 Hz
Data matrix	Time samples	Recording duration	Sampling rate

Primary objective. Before training a classifier, treat the first stage as data forensics: understand what each channel represents, how the expert annotations line up with the physiology, what normal breathing looks like, what real events look like, and what artifacts could fool a model.

1. Start with the data model: what does one row mean?

At 20 Hz, each row represents approximately 0.05 seconds of the recording. The seven columns are synchronized, so values in a row refer to approximately the same instant in the night.

time	thorax	airFlow	SpO2	arousalMean	aflowBreaths	sleepStages	OriginalDetections
0.00 s	...	...	...	...	...	...	...

A useful conceptual split is: primary physiology -> derived features / clinical context -> ground truth.

Channel guide

Channel	What it represents	Role	Key question
thorax	Thoracic respiratory effort belt signal	Primary physiology	Is the chest still trying to breathe when airflow falls?
airFlow	Filtered nasal-pressure airflow waveform	Primary physiology	Does airflow disappear, decrease, or become irregular?
SpO2	Blood oxygen saturation	Primary physiology	Is there a delayed desaturation after a respiratory change?
arousalMean	EEG-derived arousal amplitude envelope	Derived feature	Does arousal activity rise near the end of an event?
aflowBreaths	Breath-amplitude / rolling-baseline feature derived from airflow	Derived feature	How much is each breath reduced relative to recent breathing?
sleepStages	Wake/N1/N2/N3/REM state sampled at 20 Hz	Clinical context	Was the person actually asleep during the interval?
OriginalDetections	Binary expert respiratory-event mask	Ground truth	Is a scored respiratory event active at this time point?

Important: OriginalDetections is a target/ground-truth channel, not a model input at inference time.

2. Learn the visual language of normal breathing

Before examining apnea, inspect several 1-2 minute intervals where OriginalDetections is inactive. Plot all channels on a shared time axis.

airFlow: Identify the repeating breath cycles and their normal amplitude variability.

thorax: Compare respiratory effort to airflow. The waveforms do not need to match exactly, but they should show a related breathing rhythm.

Internal project working note | Based on the BioHackathon Team 14 synthesis report and sample .mat file inspection

SpO2: Expect a slower signal. Oxygen changes can lag respiratory abnormalities by several seconds.

arousalMean and aflowBreaths: Learn their normal baseline behavior before interpreting spikes or reductions during events.

3. Build an event viewer and work backward from known annotations

Use events.apneas and events.hypopneas to jump directly to expert-scored examples. Start with roughly 10 apneas and 10 hypopneas, each shown with 30-60 seconds of context before and after the event.

For every event, ask	What to inspect
A. What happened to airflow?	Near-disappearance suggests apnea-like physiology; a sustained amplitude reduction is more consistent with hypopnea-like physiology.
B. What happened to thoracic effort?	Persistent effort during absent airflow is physiologically different from simultaneous loss of airflow and effort.
C. What happened to oxygen?	Measure whether SpO2 falls, by how much, and how long after the respiratory change the nadir occurs.
D. What happened to arousalMean?	Look for an increase near the event termination or recovery period.
E. What happened to aflowBreaths?	Determine how this engineered airflow-reduction feature behaves in normal breathing, apnea, and hypopnea.

4. Verify the annotation structure

Check exactly what OriginalDetections contains. The project report describes it as a binary event mask (0 = normal; 1000 = event active). Verify this empirically with `np.unique(...)`, then confirm that intervals in events.apneas and events.hypopneas overlap the active mask as expected.

Why this matters: the binary mask appears to answer “is some respiratory event active?”, while the separate event arrays provide apnea vs hypopnea boundaries. That distinction determines how you construct training labels.

5. Reproduce sleep time and AHI yourself

The recording is 4.76 hours long, but recording duration is not necessarily total sleep time. Use sleepStages to calculate time spent in Wake, N1, N2, N3, and REM. Then calculate:

$$\text{AHI} = (\text{number of apneas} + \text{number of hypopneas}) / \text{total sleep time in hours}$$

Compare your independently calculated AHI to metrics.ahi. Agreement validates your understanding of time units, sleep-stage encoding, event arrays, and the clinical summary metric.

6. Quantify events instead of only eyeballing them

Once the waveforms make visual sense, build an event-level table. Useful columns include:

event type | event duration | airflow amplitude reduction | thorax amplitude change | pre-event SpO2 | SpO2 nadir | change in SpO2 | delay to SpO2 nadir | arousalMean peak | sleep stage

This is where qualitative waveform observations become measurable features and where apnea/hypopnea differences can be tested across many events.

7. Study difficult negatives and artifacts

A useful detector must recognize abnormal-looking signals that are not true apnea events. Deliberately search for:

- wakefulness and stage transitions
- movement/coughing
- sensor displacement or signal dropout
- flat or clipped channels
- low SpO2 without a respiratory event
- abrupt airflow changes without expert event annotation

This is especially important because the project goal includes robustness to noisy unattended recordings.

8. Characterize signal quality across each recording

For every channel, calculate min, max, median, mean, standard deviation, NaN count, fraction of constant/flat samples, and suspicious/extreme values. Plot distributions where useful.

Key interpretation issue: a flat airflow channel could indicate apnea, but it could also indicate a displaced cannula. The model ultimately needs to distinguish physiology from sensor failure.

9. Only then decide which channels should be model inputs

The processed files mix physiological measurements with engineered features. That is valid for a hackathon, but it changes what the model is learning. A useful ablation sequence is:

Model	Inputs
A	airFlow + thorax + SpO2
B	A + arousalMean
C	B + aflowBreaths
D	C + sleepStages

This reveals how much performance comes from primary physiology versus clinically engineered features.

10. Resolve the target-label question before training

The project description is slightly ambiguous about apnea subtypes. The current processed structure clearly supports at least:

- normal vs respiratory event
- normal vs apnea vs hypopnea

It is not yet clear from the report alone whether the processed files preserve obstructive vs central vs mixed apnea labels separately. Confirm this with the team before designing a multiclass target.

Recommended first three notebook sessions

Session	Goal	Concrete outputs
1. Subject summary	Understand one file end-to-end	duration; stage durations; channel ranges; missing values; counts of apnea/hypopnea; AHI check
2. Event browser	Learn event morphology	normal windows; 10 apnea examples; 10 hypopnea examples; aligned multichannel plots
3. Event table + QC	Turn observations into measurements	event-level feature table; artifact examples; signal-quality flags; first hypotheses for model inputs

Open questions to confirm with the project team

- Are obstructive, central, and mixed apnea labels preserved anywhere in the .mat files, or are they collapsed into one apnea class?
- Is the intended hackathon model allowed to use sleepStages, arousalMean, and aflowBreaths as inputs, or should some be treated as reference/derived information only?
- How were signal dropouts and artifacts encoded during preprocessing (NaN, zero, flatline, other sentinel values)?
- Will hidden evaluation data use the same preprocessing pipeline and the same channel definitions?
- How should event-boundary tolerance be handled when converting per-sample predictions to event-level detections?

Bottom line

Do not start with model tuning. First establish a trustworthy mental and quantitative model of the signals, event labels, sleep-time calculation, and artifacts. The most valuable near-term tool is a reusable subject-summary + event-browser notebook that can be run across all recordings.

Source note: The channel definitions, preprocessing description, sleep-stage encoding, event structures, and AHI definition summarized here are based on the uploaded “St. Jude BioHackathon 2026 - Team 14 - Multimodal Sleep Disordered Breathing: Challenge Synthesis & Technical Specifications” report. The sample-file dimensions and duration are from the inspected example provided by the project team.